

## Pi-Ager install on Raspberry Pi 3/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite bookworm

- **For all Pi:**

Download and install Raspberry Pi Imager v1.8.4 or later.

Start Raspberry Pi Imager, then select:

Raspberry Pi Device:

NO FILTERING

Operating System:

Raspberry Pi OS Lite (32-bit) ( A port of Debian Bookworm with no desktop environment)

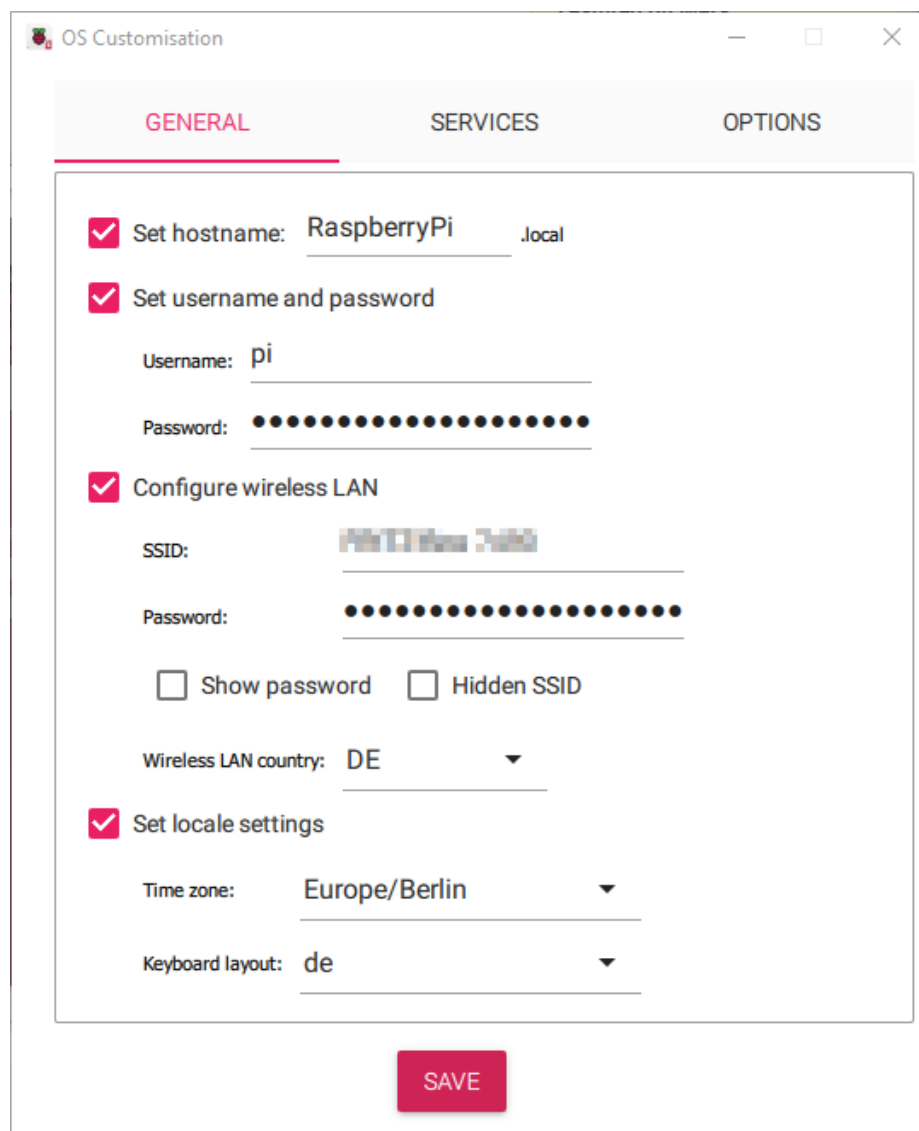
Storage:

Use an USB Reader and choose your SD-Card, min. 8 GB

Then click 'NEXT'

In popup Window 'Use OS customisation' select 'EDIT SETTINGS'.

In the 'GENERAL' Tab edit your settings:

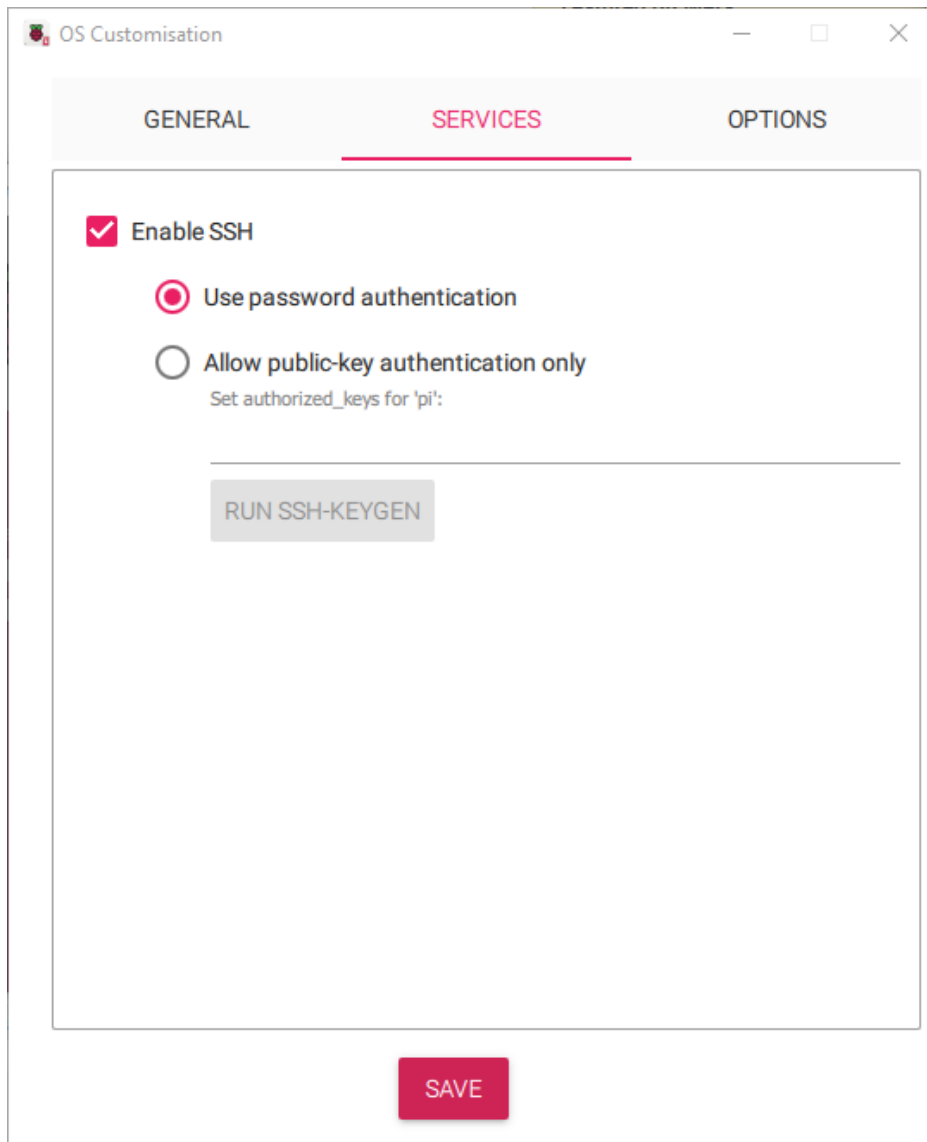


The screenshot shows the 'OS Customisation' window with the 'GENERAL' tab selected. The window has three tabs: 'GENERAL', 'SERVICES', and 'OPTIONS'. The 'GENERAL' tab contains several settings:

- ☒ Set hostname: RaspberryPi.local
- ☒ Set username and password
  - Username: pi
  - Password: [masked]
- ☒ Configure wireless LAN
  - SSID: [masked]
  - Password: [masked]
  - ☐ Show password ☐ Hidden SSID
  - Wireless LAN country: DE
- ☒ Set locale settings
  - Time zone: Europe/Berlin
  - Keyboard layout: de

A red 'SAVE' button is located at the bottom center of the window.

In the 'SERVICES' Tab enable SSH:



Then 'SAVE' and apply OS customisation settings with 'YES'.

The customized OS is now written to your SD-Card.

Put your SD-Card in your Raspberry Pi and power on your Pi device.

Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to your Raspberry device and continue with the setup as described below.

- To allow root login:  
In /etc/ssh/sshd\_config : remove # from PermitRootLogin and replace prohibit-password to yes. Then restart ssh server: `sudo service ssh restart`  
Set your root password with: `sudo passwd root`
- Run raspi-config and disable login shell on serial port and enable serial port
- Edit config.txt in /boot/firmware to support I2C and SPI devices:

**CAUTION:** If you want to edit config.txt ,cmdline.txt and setup.txt under Windows in a Terminal Window, e.g. using Terminal Apps like PuTTY, the Linux folder /boot/firmware on

the SD-Card is mapped to a windows partition FAT32 USB Drive named bootfs and contains the above mentioned files.

```
# Additional overlays and parameters are documented /boot/overlays/README
# Use Pi-Ager Pins 11/13 GPIO 17/27 for I2C
dtoverlay=i2c-gpio,bus=3,i2c_gpio_sda=17,i2c_gpio_scl=27
# Use Pi-Ager Pin 16 for MCP3204
dtoverlay=spi1-1cs,cs0_pin=16
```

at the end of config.txt add the following lines to support bluetooth and Nextion TFT displays via serial port /dev/serial0 :

```
[all]
enable_uart=1
dtoverlay=miniuart-bt
dtparam=uart0=on
# force_turbo=1
```

- Add in /boot/firmware/cmdline.txt at the end of line this to enable USB camera with fswebcam :  
`fsck.mode=force dwc_otg.fiq_fsm_mask=0x3`
- Reboot system
- Edit /etc/modules to load i2c-dev at boot, add this line :  
`i2c-dev`
- Add file :  
`sudo touch /etc/modprobe.d/raspi-blacklist.conf`
- Install git: `sudo apt install git`
- Get a copy from Pi-Ager repository to your local system:  
`git clone --depth=1 -b master https://github.com/Tronje-the-Falconer/Pi-Ager`  
All project file are now in the folder ./Pi-Ager/  
Change working directory to Pi-Ager  
`cd Pi-Ager`
- Copy setup.txt from local repository to /boot/ and edit it as needed:  
`sudo cp ./boot/firmware/setup.txt /boot/firmware/`
- Reboot system
- Update system:  
`sudo apt-get update`  
`sudo apt-get upgrade`
- Install php 8:  
`sudo apt install php-fpm php-cli`  
`sudo apt install php`  
`sudo apt install php-common php-sqlite3`
- Install additional modules for php:

```
sudo apt install php-mbstring php-zip php-curl
```

- Install lighttpd:

```
sudo apt-get install lighttpd
sudo systemctl status lighttpd
```

```
sudo nano /etc/lighttpd/lighttpd.conf
and change/add Parameter
```

```
server.document-root = "/var/www/html"
to
server.document-root = "/var/www"
add "mod_auth" to server.modules list
add "mod_authn_file" to server.modules list
Save and close nano
```

```
/*----- not used -----*/
```

To enable PHP8 in Lighttpd, we must modify `/etc/php/8.2/fpm/php.ini` and uncomment the line `cgi.fix_pathinfo=1`. In my `php.ini` file, I found it on line 807. To uncomment, just remove the semicolon in the beginning.

```
/*-----*/
```

```
sudo usermod -G www-data -a pi
sudo chown -R www-data:www-data /var/www
sudo chmod -R 755 /var/www
```

- Reboot system

For testing the web server, generate html-page:

```
sudo nano /var/www/test.html
with content:
```

```
<html>
<head><title>Test-Seite</title></head>
<body>
<h1>Das ist eine Test-Seite.</h1>
</body>
</html>
```

Change owner: `sudo chown www-data:www-data /var/www/test.html`

Enter your IP Address into the browser followed by `/test.html`

In addition we need `.htcredentials` to contain user and password.

For that we use the Online-Tool <https://websistent.com/tools/htdigest-generator-tool/>

Username: pi-ager

REALM: Pi-Ager

Password: raspberry

Caution! All entries are case sensitive!

Open this file now

```
sudo nano /var/.htcredentials
```

and fill in the string output from the generator tool.

Save file with “STRG+o”, “RETURN” and close with “STRG+x”

Now we have to setup the password authentication in lighttpd:

```
sudo nano /etc/lighttpd/conf-available/05-auth.conf
```

The following lines are added under `server.modules += („mod_auth“)` :

```
auth.backend          = "htdigest"
auth.backend.htdigest.userfile = "/var/.htcredentials"
```

```
auth.require          = ( "/settings.php" =>
    (
        "method" => "digest",
        "realm" => "Pi-Ager",
        "require" => "user=pi-ager"
    ),
    "/admin.php" =>
    (
        "method" => "digest",
        "realm" => "Pi-Ager",
        "require" => "valid-user"
    ),
    "/webcam.php" =>
    (
        "method" => "digest",
        "realm" => "Pi-Ager",
        "require" => "valid-user"
    ),
    "/notification.php" =>
    (
        "method" => "digest",
        "realm" => "Pi-Ager",
        "require" => "valid-user"
    )
)
```

Then we activate this modul:

```
sudo lighty-enable-mod auth
```

In addition we have to edit :

```
sudo nano /etc/lighttpd/conf-available/15-fastcgi-php-fpm.conf
```

Add “broken-scriptfilename” and “x-sendfile” :

```
fastcgi.server += ( ".php" =>
    (
        "socket" => "/run/php/php-fpm.sock",
```

```
        "broken-scriptfilename" => "enable",
        "x-sendfile" => "enable"
    ))
)
```

save and close nano

Now enable these modules:

```
sudo lighty-enable-mod fastcgi-php-fpm
```

Now reload the the webserver:

```
sudo service lighttpd force-reload
```

Now continue to install additional modules:

- Install smbush  

```
sudo apt-get install python3-smbus2
```
- Install sqlite3:  

```
sudo apt install sqlite3
```
- Install pip3  

```
sudo apt install python3-pip
```

**CAUTION :**

```
sudo pip3 install <module>
```

 no longer supported.

If you see an error message, try to install the module with  

```
sudo apt install python3-<module name>
```

. If this does not work, try install modules with  

```
sudo pip3 install <module name> --break-system-packages
```

```
error: externally-managed-environment
× This environment is externally managed
└─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.
```

If you wish to install a non-Debian-packaged Python package,  
create a virtual environment using `python3 -m venv path/to/venv`.  
Then use `path/to/venv/bin/python` and `path/to/venv/bin/pip`. Make  
sure you have `python3-full` installed.

If you wish to install a non-Debian packaged Python application,  
it may be easiest to use `pipx install xyz`, which will manage a  
virtual environment for you. Make sure you have `pipx` installed.

See `/usr/share/doc/python3.11/README.venv` for more information.  
note: If you believe this is a mistake, please contact your Python  
installation or OS distribution provider. You can override this, at the  
risk of breaking your Python installation or OS, by passing `--break-system-  
packages`.  
hint: See PEP 668 for the detailed specification.

- Install SHT1x sensors  
`sudo pip3 install pi-sht1x --break-system-packages`
- Install libgd-dev (needed for new version of fswebcam)  
`sudo apt install libgd-dev`
- Install openssl dev package  
`sudo apt install libssl-dev`
- Install fswebcam:  
`sudo apt install fswebcam`
- Install cryptography  
`sudo apt install python3-cryptography`
- Install uuidgen  
`sudo apt install uuid-runtime`
- Install wiringpi new version with Pi 5 support :  
`cd /tmp`  
`sudo apt purge wiringpi`  
`wget`  
[https://github.com/phylax2020/WiringPi/releases/download/3.16/wiringpi\\_3.16\\_armhf.deb](https://github.com/phylax2020/WiringPi/releases/download/3.16/wiringpi_3.16_armhf.deb)  
`sudo apt install ./wiringpi_3.16_armhf.deb`
- Install RPi.GPIO new version with support for RPi 5  
`cd /tmp`  
`sudo apt purge python3-RPi.GPIO` or  
`sudo pip3 uninstall RPi.GPIO --break-system-packages`  
`wget` [https://github.com/phylax2020/RPi.GPIO/releases/download/v0.8.6/python3-rpi.gpio\\_0.8.6-1\\_armhf.deb](https://github.com/phylax2020/RPi.GPIO/releases/download/v0.8.6/python3-rpi.gpio_0.8.6-1_armhf.deb)  
`sudo apt install ./python3-rpi.gpio_0.8.6-1_armhf.deb`
- Nextion serial client (HMI Display support)  
`sudo pip3 install nextion --break-system-packages`  
`sudo pip3 install pyserial-asyncio --break-system-packages`
- Install lsof command:  
`sudo apt install lsof`
- Install Locale en-GB and de-DE UTF-8 using  
`sudo raspi-config`
- Enable Serial Interface, disable login, needed for HMI Nextion Display  
`sudo raspi-config`
- Install zip and unzip:  
`sudo apt install zip unzip`
- Install schedule

`sudo apt install python3-schedule`

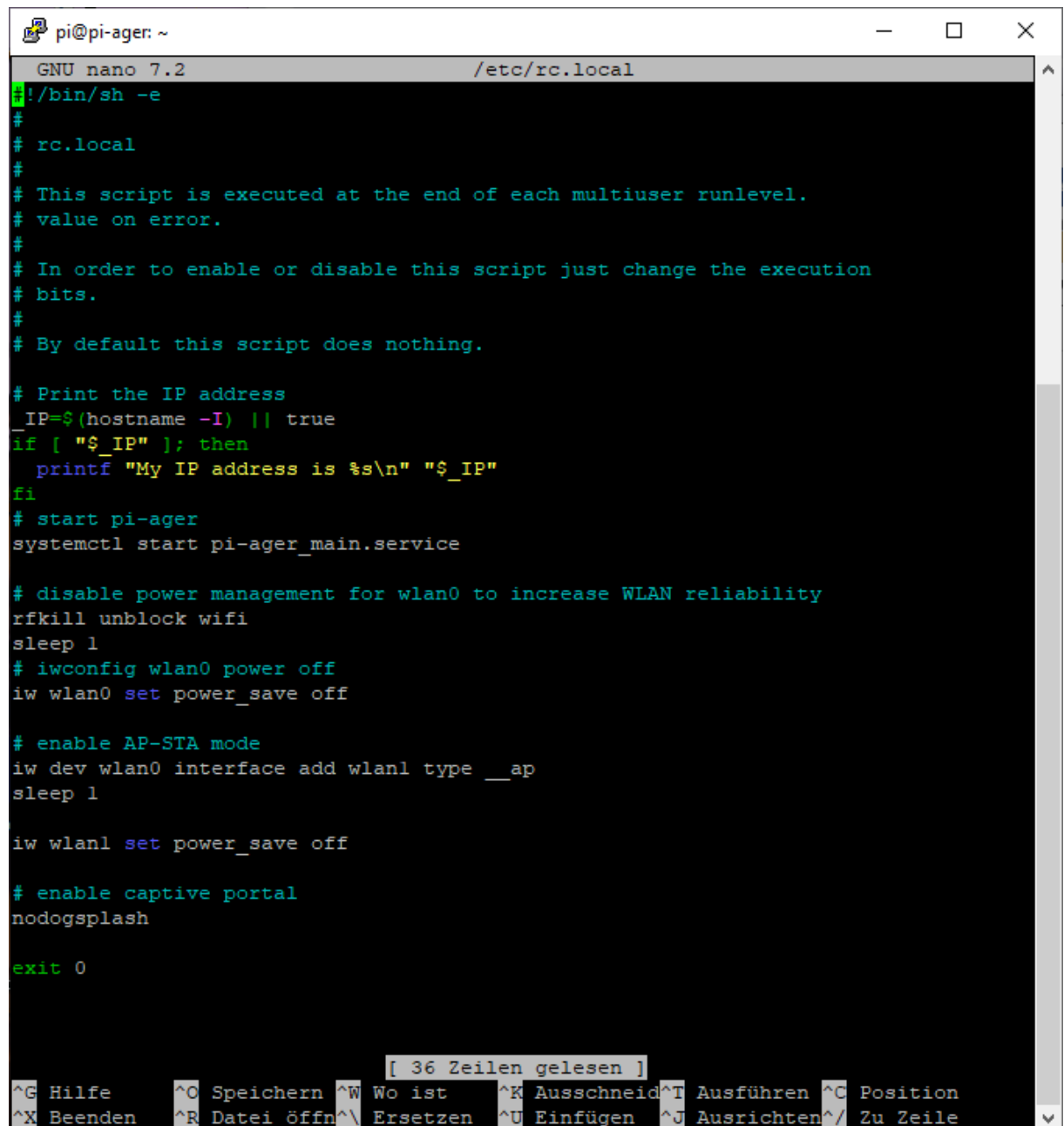
- Install MQTT client

`sudo apt install python3-paho-mqtt`

- Unblock wifi for Pi4, add rfkill unblock wifi and disable power management for wlan0:

`cd /etc`

`sudo nano rc.local`



The screenshot shows a terminal window titled 'pi@pi-ager: ~' with a window manager title bar. The terminal is running GNU nano 7.2 editing the file /etc/rc.local. The content of the file is as follows:

```
#!/bin/sh -e
#
# rc.local
#
# This script is executed at the end of each multiuser runlevel.
# value on error.
#
# In order to enable or disable this script just change the execution
# bits.
#
# By default this script does nothing.

# Print the IP address
_IP=$(hostname -I) || true
if [ "$_IP" ]; then
    printf "My IP address is %s\n" "$_IP"
fi
# start pi-ager
systemctl start pi-ager_main.service

# disable power management for wlan0 to increase WLAN reliability
rfkill unblock wifi
sleep 1
# iwconfig wlan0 power off
iw wlan0 set power_save off

# enable AP-STA mode
iw dev wlan0 interface add wlan1 type __ap
sleep 1

iw wlan1 set power_save off

# enable captive portal
nodogsplash

exit 0
```

At the bottom of the terminal, a status bar indicates '[ 36 Zeilen gelesen ]' and provides navigation shortcuts: ^G Hilfe, ^O Speichern, ^W Wo ist, ^K Ausschneid, ^T Ausführen, ^C Position, ^X Beenden, ^R Datei öffn, ^\ Ersetzen, ^U Einfügen, ^J Ausrichten, ^/ Zu Zeile.

Then change protection and owner:

`sudo chown root:root /etc/rc.local`

`sudo chmod +x /etc/rc.local`

- Only if wanted: Generate/edit crontab to prepare for automatic enable pi-ager\_backup.sh

```
pi@pi-ager:~$ sudo crontab -l
#-----
# Shell variable for cron
SHELL=/bin/bash
# PATH variable for cron
PATH=/usr/local/bin:/usr/local/sbin:/sbin:/usr/sbin:/bin:/usr/bin:/usr/bin/X11
#M S T M W Befehl
#-----
#* */6 * * * /usr/bin/flock -w 0 /var/run/pi-ager_backup.pid /usr/local/bin/pi-ager_backup.sh >> /var/log/pi-ager_backup
.log 2>&1
#*/2 * * * * logger "greetings from the crontab" > /dev/null 2>&1
```

- Use visudo to edit /etc/sudoers, so that the www-data User (User of Website) can execute /var/sudowebscript.sh :

`sudo visudo`

and then in sudoers following

...

#User privilege specification

root ALL=(ALL:ALL) ALL

...

adding:

`www-data ALL=NOPASSWD:/var/sudowebscript.sh, /var/show_wifi_connections.sh, /var/updatessid.sh,/usr/bin/raspi-config,/usr/bin/nmcli`

Save and exit.

- Install access point with network manager:  
Copy from repository /usr/local/bin/setup\_pi-ager-ap.sh to destination.

Create virtual interface wlan1:

`sudo iw dev wlan0 interface add wlan1 type __ap`

then run script :

`cd /usr/local/bin`

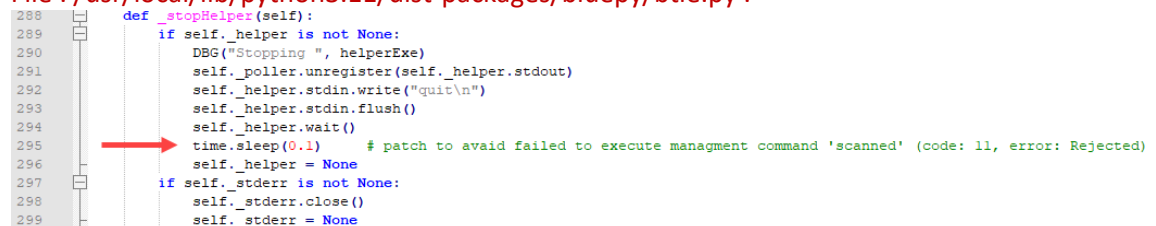
`sudo ./setup_pi-ager-ap.sh`

- Install nodogsplash captive portal  
`sudo apt install iptables`  
`sudo apt install libmicrohttpd-dev`  
`cd ~`  
`git clone https://github.com/nodogsplash/nodogsplash.git`  
`cd ./nodogsplash`  
`make`  
`sudo make install`  
copy all from repository /etc/nodogsplash to /etc/nodogsplash  
`sudo chmod +x /usr/bin/nodogsplash`  
`sudo chmod +x /usr/bin/ndsctl`

- Install Bluetooth modules to support Bluetooth Temp./Hum Sensor from Xiaomi

- `sudo apt install libglib2.0-dev`
- `sudo pip3 install bluepy requests --break-system-packages`
- `sudo apt install bluetooth libbluetooth-dev`
- `# sudo pip3 install pybluez pycryptodomex`
- Patch btle.py to stop intermittend errors:

File : /usr/local/lib/python3.11/dist-packages/bluepy/btle.py :



```

288 def _stopHelper(self):
289     if self._helper is not None:
290         DBG("Stopping ", helperExe)
291         self._poller.unregister(self._helper.stdout)
292         self._helper.stdin.write("quit\n")
293         self._helper.stdin.flush()
294         self._helper.wait()
295         time.sleep(0.1) # patch to avoid failed to execute managment command 'scanned' (code: 11, error: Rejected)
296         self._helper = None
297     if self._stderr is not None:
298         self._stderr.close()
299         self._stderr = None

```

- Now copy all files and folders from your local git repository /var/www to /var/www/
- from local repository /opt/pi-ager/ to /opt/pi-ager/

- from local repository /var/sudowebscript.sh to /var/
- `sudo chown -R www-data:www-data /var/www`
- `sudo usermod -G gpio -a www-data`
- `sudo chmod 666 /var/www/logs/logfile.txt`
- `sudo chmod 755 /var/www/logs/`
- `sudo chmod 664 /var/www/config/pi-ager.sqlite3`
- `sudo chown -R www-data:www-data /var/www/config/`
- `sudo chmod 777 /var/www/config/`
- `sudo chmod 555 /var/sudowebscript.sh /var/updatessid.sh /var/show_wifi_connections.sh`
- from local repository /usr/local/bin/\*.sh copy all to /usr/local/bin/ (pi-ager\_backup.sh, pi-ager\_image.sh, setup\_pi-ager.sh)  
Set +x mode to the scripts :  
`sudo chmod +x /usr/local/bin/*.sh`
- Bluetooth sensor: Make ATC\_XXXXXX.py executable  
`sudo chmod +x /opt/ATC_MiThermometer/ATC_XXXXXX.py`
- from local repository /lib/systemd/system copy the following files to /lib/systemd/system/ :  
`pi-ager_main.service`  
`setup_pi-ager.service`  
Then reload these services:  
`sudo systemctl daemon-reload`
- from local repository /usr/bin copy the following file to /usr/bin/. This is a newer version of fswebcam with re-get frame on error.  
`fswebcam`  
Set +x mode to fswebcam:  
`sudo chmod +x /usr/bin/fswebcam`
- from local repository /usr/share/man/man1/fswebcam.1.gz copy the following file to /usr/share/man/man1/  
`fswebcam.1.gz`
- enable setup\_pi-ager.service to start only once after reboot  
`sudo systemctl enable setup_pi-ager.service`
- Restart system  
`sudo reboot`

## How to generate and deploy Pi-Ager Images

There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.
2. On the NFS Server System Install Pi Power Tools using the well-known Pi-Apps.
3. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :

## BACKUP

|                  |                                                  |
|------------------|--------------------------------------------------|
| nfs Verzeichnis: | 192.168.0.238:/home/public/pi-ager-v400-noshrink |
| Anzahl Backups:  | 3                                                |
| Backup Name:     | PiAgerBackup                                     |
| nfs Option:      | nosuid,nodev                                     |
| Backup Aktiv:    | <input checked="" type="checkbox"/>              |

Speichern  
Backup manuell  
pi-ager\_backup.log in einem neuen Tab öffnen

Hilfe

4. Next start a terminal on the Pi-Ager and start the pi-ager\_image.sh script with option -l:

```

pi@pi-ager: /usr/local/bin
pi@pi-ager:~ $ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin $ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy = false
my image = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
#####
parted output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
#####
parted_output boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
#####
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service -> /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin $

```

5. Zip and deploy your new image.